

Challenge Lab: Build Your DB Server and Interact with Your DB

Created by: Kevin Carl B. Ricafort

Cohort: PHMAN29

This lab is designed to reinforce the concept of leveraging an AWS-managed database instance for solving relational database needs.

Amazon Relational Database Service (Amazon RDS) makes it easy to set up, operate, and scale a relational database in the cloud. It provides cost-efficient and resizable capacity while managing time-consuming database administration tasks, which allows you to focus on your applications and business. Amazon RDS provides you with six familiar database engines to choose from: Amazon Aurora, Oracle, Microsoft SQL Server, PostgreSQL, MySQL and MariaDB. This lab is designed to reinforce the concept of leveraging an AWS-managed database instance for solving relational database needs.

Amazon Relational Database Service (Amazon RDS) makes it easy to set up, operate, and scale a relational database in the cloud. It provides cost-efficient and resizable capacity while managing time-consuming database administration tasks, which allows you to focus on your applications and business. Amazon RDS provides you with six familiar database engines to choose from: Amazon Aurora, Oracle, Microsoft SQL Server, PostgreSQL, MySQL and MariaDB.

Initial Configurations

Before creating and interacting the database, Amazon Aurora/RDS configurations must be ensured. Take note that some services and functionalities are initially configured such as EC2 instances, VPC, and security group (at least for this laboratory activity).

Key	Value
Database Engine	Used supported engines like Amazon Aurora or MySQL.
Template	Dev/Test or Free Tier.
Availability and Durability	Do not create standby instance.
DB Instance Size	I utilized a Burstable class <code>db.t*.small</code> to avoid underutilized resources and higher costings.
Amazon VPC	I utilized the default VPC named <code>Lab VPC</code>
Security Group	I created another security group that has three inbound groups <code>MySQL/Aurora, MSSQL, HTTPS, SSH</code>
Storage	General Purpose SSD (gp2) with a 100 GB Size
Master User Name	maindb
Password	lab-password

Changing to root

```
sudo su -
```

Installing MariaDB Server

```
yum install -y mariadb-server
```

Start and Enable the MariaDB Service

```
systemctl enable --now mariadb
```

Verify if it is running

```
systemctl status mariadb
```

Connect to MariaDB

```
mysql -u root
```

Challenge Proper

Part 1: Create a table named RESTART with the following columns:

- Student ID (Number)
- Student Name
- Restart City
- Graduation Date (Graduation Time)

```
MariaDB [test]> SELECT * FROM test.RESTART;
```

Student_ID	Student_Name	Restart_City	Graduation_Date
AWS-25-1001	Liam Vance	Cambridge	2025-05-28
AWS-25-1002	Sophia Alcantara	San Jose	2025-12-28
AWS-25-1003	Mateo Hernandez	Austin	2025-05-28
AWS-25-1004	Chloe Davenport	Seattle	2025-05-27
AWS-25-1005	Ethan Takahashi	Boston	2025-12-15
AWS-25-1006	Darwin Fischer	Chicago	2025-05-28
AWS-25-1007	Maya Lin	Denver	2025-12-15
AWS-25-1008	Zara Patel	San Diego	2025-12-18
AWS-25-1009	Noah Sterling	Atlanta	2025-05-28
AWS-25-1010	Alexander Wright	Cambridge	2025-06-26

```
10 rows in set (0.00 sec)
```

Figure 1.1. The output table with the four required columns from the RESTART table.

Part 2: Create a table named CLOUD_PRACTITIONER with the following columns:

- Student ID(Number)
- Certification Date (Date Time)

```
MariaDB [test]> SELECT * FROM CLOUD_PRACTITIONER;
```

Student_ID	Certification_Date
AWS-25-1001	2025-06-28
AWS-25-1003	2026-01-17
AWS-25-1004	2025-06-28
AWS-25-1007	2026-01-17
AWS-25-1010	2025-06-28

```
5 rows in set (0.00 sec)
```

Figure 2.1. The output table with the two required columns from the CLOUD_PRACTITIONER table.

Part 3: Perform an inner join between 2 tables created and display student ID, Student Name, Certification Date

```
MariaDB [test]> SELECT r.Student_ID, r.Student_Name, c.Certification_Date  
-> FROM RESTART r INNER JOIN CLOUD_PRACTITIONER c  
-> ON r.Student_ID = c.Student_ID;
```

Student_ID	Student_Name	Certification_Date
AWS-25-1001	Liam Vance	2025-06-28
AWS-25-1003	Mateo Hernandez	2026-01-17
AWS-25-1004	Chloe Davenport	2025-06-28
AWS-25-1007	Maya Lin	2026-01-17
AWS-25-1010	Alexander Wright	2025-06-28

5 rows in set (0.00 sec)

Figure 3.1. This table is the output after the user performed an INNER JOIN between the two tables.